

The Comprehensive Clinical Master Handbook for MBBS & NEET-PG

Focus: High-Yield Internal Medicine (Top 10 Core Topics)

1. DIABETES MELLITUS (DM)

Definition

A metabolic disorder characterized by chronic hyperglycemia resulting from defects in insulin secretion, insulin action, or both, leading to disturbances in carbohydrate, fat, and protein metabolism.

Etiology

- **Type 1 DM:** Autoimmune destruction of pancreatic β -cells (Type IV hypersensitivity), leading to absolute insulin deficiency. Associated with HLA-DR3 and HLA-DR4.
- **Type 2 DM:** Progressive insulin secretory defect on the background of insulin resistance. Associated with obesity, genetics (polygenic), and sedentary lifestyle.
- **Other Specific Types:** Genetic defects of β -cell function (MODY), pancreatic exocrine disease (chronic pancreatitis, cystic fibrosis), endocrinopathies (Cushing's, Acromegaly), drug-induced (Corticosteroids, Thiazides).
- **Gestational DM:** Onset or first recognition during pregnancy.

Pathophysiology

- **Type 1:** Genetic predisposition + environmental trigger (e.g., Coxsackie B virus) $\rightarrow$ Autoantibodies (Anti-GAD65, ICA, IAA) $\rightarrow$ β -cell destruction $\rightarrow$ Loss of insulin $\rightarrow$ Uninhibited gluconeogenesis/glycogenolysis and lipolysis $\rightarrow$ Hyperglycemia and Ketoacidosis.
- **Type 2:** Visceral obesity/adipokines cause peripheral insulin resistance (decreased GLUT-4 translocation in muscle/fat) $\rightarrow$ Compensatory hyperinsulinemia $\rightarrow$ Eventual β -cell exhaustion and apoptosis (amylin amyloid deposition) $\rightarrow$ Hyperglycemia. Incretin effect (GLP-1/GIP) is also blunted.

Clinical Features

- **Classic Triad:** Polyuria (osmotic diuresis), Polydipsia (thirst from hyperosmolarity), Polyphagia.
- **Other:** Unexplained weight loss (Type 1), blurred vision, fatigue, delayed wound healing, recurrent infections (candidiasis, UTI).
- **Acanthosis Nigricans:** Velvety hyperpigmentation in skin folds (marker of insulin resistance).

Diagnosis & Investigations

- **2-hour Post-Prandial (75g OGTT):** ≥ 200 mg/dL.
- **HbA1c:** $\geq 6.5\%$ (must be standardized to DCCT assay).
- **Random Plasma Glucose:** ≥ 200 mg/dL with classic symptoms of hyperglycemia or crisis.
- *Note: In the absence of unequivocal hyperglycemia, results require two abnormal test results from the same sample or in two separate test samples.*
- **Investigations:** Lipid profile, Serum creatinine, eGFR, Urine albumin-to-creatinine ratio (UACR), Fundoscopy, ECG. Autoantibodies (Type 1), C-peptide (low in Type 1, high/normal in Type 2).

Treatment

- **Type 1 DM:** Lifelong Insulin therapy (Basal-Bolus regimen is physiological).
- **Type 2 DM:**
 - *Lifestyle:* Diet modification, exercise (150 min/week), weight loss.
 - *First-line:* **Metformin** (decreases hepatic gluconeogenesis; avoid if eGFR < 30).
 - *Add-on therapy (based on comorbidities):*
 - **ASCVD/Heart Failure/CKD:** SGLT2 inhibitors (Empagliflozin, Dapagliflozin) or GLP-1 RAs (Liraglutide, Semaglutide). SGLT2i provide mortality benefit in HF/CKD.
 - **High cost/Access issues:** Sulfonylureas (Glimepiride - risk of hypoglycemia/weight gain), Pioglitazone (risk of edema/HF), DPP-4 inhibitors (Gliptins - weight neutral).
- **Targets:** HbA1c $< 7.0\%$ for most non-pregnant adults.

Complications

- **Acute:** Diabetic Ketoacidosis (DKA - mostly Type 1), Hyperosmolar Hyperglycemic State (HHS - mostly Type 2), Hypoglycemia.
- **Chronic Microvascular:** Retinopathy (Non-proliferative vs. Proliferative), Nephropathy (Kimmelstiel-Wilson nodules), Neuropathy (Stocking-glove sensorimotor, Autonomic).
- **Chronic Macrovascular:** Coronary Artery Disease (CAD), Peripheral Arterial Disease (PAD), Cerebrovascular Disease (Stroke).

Important Exam Points (NEET-PG)

- **MCQ Pearl:** Most common cause of death in DM type 2 is cardiovascular disease (Myocardial Infarction).
- **Pathology:** Kimmelstiel-Wilson (KW) lesions are nodular glomerulosclerosis, pathognomonic for diabetic nephropathy. Stain PAS positive.
- **DKA Triad:** Hyperglycemia (> 250 mg/dL), Ketosis, High Anion Gap Metabolic Acidosis (pH < 7.3 , Bicarb < 18).
- **Treatment of DKA:** IV Fluids (most critical initial step - 0.9% NS) $\rightarrow$ IV Regular Insulin $\rightarrow$ Potassium correction (Insulin drives K^+ into cells).

euglycemic DKA and GU infections.

2. HYPERTENSION (HTN)

Definition

Sustained elevation of systemic arterial blood pressure. According to ACC/AHA (2017), HTN is defined as Systolic BP ≥ 130 mmHg OR Diastolic BP ≥ 80 mmHg. (Note: Many global guidelines, including JNC 8 and ESC/ESH, still use $\geq 140/90$ mmHg).

Etiology

- **Primary (Essential) HTN (90-95%):** Idiopathic. Complex interplay of genetics, sympathetic nervous system (SNS) hyperactivity, Renin-Angiotensin-Aldosterone System (RAAS) overactivity, defect in natriuresis, high sodium intake, obesity, and alcohol.
- **Secondary HTN (5-10%):**
 - *Renal:* Chronic kidney disease, Renal artery stenosis (fibromuscular dysplasia in young females, atherosclerosis in elderly).
 - *Endocrine:* Primary Hyperaldosteronism (Conn's syndrome), Pheochromocytoma, Cushing's syndrome, Acromegaly, Thyroid/Parathyroid disorders.
 - *Vascular:* Coarctation of the aorta.
 - *Drugs:* OCPs, NSAIDs, Corticosteroids, Sympathomimetics.
 - *Other:* Obstructive Sleep Apnea (OSA).

Pathophysiology

Blood Pressure = Cardiac Output (CO) $\times$ Systemic Vascular Resistance (SVR).

- Increased CO (increased fluid volume via excess sodium or RAAS, or increased sympathetic drive).
- Increased SVR (vasoconstriction via Angiotensin II, catecholamines, endothelial dysfunction/decreased Nitric Oxide, vascular remodeling/hypertrophy).

Clinical Features

- Usually asymptomatic ("Silent Killer").
- May present with morning occipital headache, dizziness, tinnitus, or palpitations.
- **Secondary HTN clues:** Abdominal bruit (Renal Artery Stenosis), delayed radio-femoral delay (Coarctation), episodic sweating/headache/palpitations (Pheochromocytoma), hypokalemia (Primary hyperaldosteronism).

Diagnosis & Investigations

- **Diagnosis:** Requires at least two elevated readings on two separate occasions using proper technique. Ambulatory BP Monitoring (ABPM) is the gold standard to rule out "White Coat" and "Masked" HTN.

- ECG (Left ventricular hypertrophy).
- Urinalysis (proteinuria), Serum Creatinine, BUN, eGFR, Electrolytes (Na^+ , K^+ , Ca^{2+}).
- Fasting lipid profile, Fasting glucose.
- Fundoscopy (Keith-Wagener-Barker grading: I - Tortuosity/Silver wiring, II - AV nicking, III - Flame hemorrhages/Cotton wool spots, IV - Papilledema).

Treatment

- **Lifestyle Modifications:** DASH diet (Dietary Approaches to Stop Hypertension), Sodium restriction (< 1.5 g/day), weight loss (most effective, ~ 1 mmHg drop per 1 kg lost), aerobic exercise, moderation of alcohol.
- **Pharmacotherapy (First-line agents):**
 1. **Thiazide diuretics:** Chlorthalidone, Hydrochlorothiazide.
 2. **Calcium Channel Blockers (CCB):** Amlodipine, Nifedipine (Dihydropyridines).
 3. **ACE Inhibitors / ARBs:** Enalapril, Ramipril / Losartan, Telmisartan. (Compelling indication in DM, CKD, HF).
- *Note: Black patients respond better to Thiazides or CCBs. Do not combine ACEi and ARB.*

Complications

- **Heart:** LVH, Heart Failure (HFrEF and HFpEF), CAD, Atrial Fibrillation.
- **Brain:** Ischemic stroke, Intracerebral hemorrhage (Charcot-Bouchard microaneurysms), Vascular dementia.
- **Kidney:** Hypertensive nephrosclerosis (benign and malignant).
- **Vessels:** Aortic dissection, Peripheral arterial disease.

Important Exam Points (NEET-PG)

- **Hypertensive Emergency:** BP $> 180/120$ WITH acute target organ damage (Encephalopathy, acute MI, pulmonary edema, aortic dissection). Treat with IV medications (Nitroprusside, Labetalol, Nicardipine). Goal: Reduce MAP by max 25% in the first hour.
- **Hypertensive Urgency:** BP $> 180/120$ WITHOUT target organ damage. Treat with oral meds.
- **Contraindications:** ACEi/ARBs are strictly contraindicated in bilateral renal artery stenosis and pregnancy.
- **Aortic Dissection BP Control:** IV Beta-blocker (Esmolol, Labetalol) is first line to reduce dP/dt (shear stress), followed by a vasodilator.

3. BRONCHIAL ASTHMA

Definition

respiratory symptoms (wheeze, shortness of breath, chest tightness, and cough) that vary over time and in intensity, together with **variable expiratory airflow limitation**.

Etiology

- **Atopic (Extrinsic):** Type I Hypersensitivity reaction to environmental allergens (dust mites, pollen, pet dander). IgE mediated. Begins in childhood, associated with positive family history and other atopic diseases (eczema, allergic rhinitis).
- **Non-Atopic (Intrinsic):** Triggered by respiratory infections (especially viral), cold air, exercise, stress, irritating fumes. Normal IgE levels.
- **Drug-Induced:** Aspirin-exacerbated respiratory disease (AERD) - Samter's Triad (Asthma, Aspirin sensitivity, Nasal polyps). Due to inhibition of COX, leading to overproduction of leukotrienes.
- **Occupational:** Exposure to sensitizing agents at work (e.g., isocyanates, wood dust).

Pathophysiology

Exposure to trigger → Activation of Th2 cells → Release of cytokines (IL-4 for IgE class switching, IL-5 for eosinophil recruitment, IL-13 for mucus hypersecretion) → IgE binds to mast cells → Cross-linking by allergen → Degranulation (Histamine, Leukotrienes, Prostaglandins).

- *Early Phase (Minutes):* Bronchoconstriction, edema, increased mucus.
- *Late Phase (Hours):* Infiltration by eosinophils, neutrophils, T-cells causing epithelial damage and hyperreactivity.
- *Airway Remodeling (Chronic):* Smooth muscle hypertrophy, subepithelial fibrosis, mucus gland hyperplasia.

Clinical Features

- Episodic breathlessness, wheezing, cough (often nocturnal or early morning), chest tightness.
- Physical Exam: Prolonged expiratory phase, polyphonic widespread expiratory wheeze, tachypnea, tachycardia. Pulsus paradoxus in severe attacks.

Diagnosis & Investigations

- **Pulmonary Function Tests (Spirometry):** Obstructive pattern. $FEV_1/FVC < 0.7$ (or lower limit of normal).
- **Reversibility Testing:** Increase in FEV_1 by $\geq 12\%$ AND ≥ 200 mL after administering a short-acting bronchodilator (SABA like Salbutamol).
- **Peak Expiratory Flow Rate (PEFR):** Diurnal variability $> 10\%$ is diagnostic.
- **Bronchoprovocation Test:** Methacholine challenge (if spirometry is normal but asthma is suspected). PC_{20} (provocative concentration causing 20% drop in FEV_1) < 8 mg/mL is positive.
- **Sputum:** Curschmann spirals (whorled mucus plugs) and Charcot-Leyden crystals (breakdown products of eosinophils - galectin-10).

- **Relievers:** Low-dose Inhaled Corticosteroid (ICS) + Formoterol (a LABA with fast onset). This is the preferred reliever in modern guidelines (MART therapy). Alternatively, SABA (Salbutamol) as needed.
- **Controllers (Stepwise):**
 - Step 1/2: As-needed low-dose ICS-Formoterol.
 - Step 3: Low-dose maintenance ICS-LABA.
 - Step 4: Medium-dose maintenance ICS-LABA.
 - Step 5: High-dose ICS-LABA + add-on (LAMA e.g., Tiotropium) or Biologics (Omalizumab [anti-IgE], Mepolizumab [anti-IL5]).
- **Acute Exacerbation (Status Asthmaticus):** Oxygen, inhaled high-dose SABA + Ipratropium, systemic corticosteroids (Oral Prednisolone or IV Hydrocortisone). IV Magnesium Sulfate in severe cases.

Complications

- Acute severe asthma (Status Asthmaticus), respiratory failure, pneumothorax, cor pulmonale (rare compared to COPD).

Important Exam Points (NEET-PG)

- **Normal or high $PaCO_2$ in an acute asthma attack:** Indicates impending respiratory failure (muscle fatigue). The patient is usually hypocapnic (respiratory alkalosis) due to tachypnea.
- **Omalizumab:** Monoclonal antibody against Fc portion of IgE. Used in severe allergic asthma.
- **Mepolizumab/Reslizumab:** IL-5 antagonists, used in severe eosinophilic asthma.
- **Montelukast/Zafirlukast:** Leukotriene receptor antagonists. Excellent for Aspirin-induced and exercise-induced asthma.

4. CHRONIC OBSTRUCTIVE PULMONARY DISEASE (COPD)

Definition

A common, preventable, and treatable disease characterized by persistent respiratory symptoms and **irreversible (or poorly reversible) airflow limitation** that is due to airway and/or alveolar abnormalities usually caused by significant exposure to noxious particles or gases.

Etiology

- **Tobacco Smoking:** Most common cause (> 90% in developed countries).
- **Biomass Fuel Exposure:** Major cause in developing countries, especially among women cooking indoors.
- **Occupational Dusts/Chemicals:** Coal miners, silica.
- **Genetic:** Alpha-1 Antitrypsin (α_1 -AT) deficiency (causes panacinar emphysema, often lower lobes, in young non-smokers or early in smokers).

Two main components often coexist:

1. **Chronic Bronchitis (Airway Disease):** Clinically defined as a productive cough for > 3 months/year for 2 consecutive years. Hypertrophy and hyperplasia of mucus-secreting glands in bronchi. *Reid Index* (thickness of mucosal gland layer / thickness of wall between epithelium and cartilage) > 0.4 .
2. **Emphysema (Alveolar Disease):** Pathologically defined as permanent enlargement of air spaces distal to terminal bronchioles with destruction of alveolar walls, without obvious fibrosis. Neutrophils/macrophages release elastases; deficiency of anti-elastases (α_1 -AT) leads to destruction of elastic tissue. Decreased radial traction causes airway collapse during expiration. Centriacinar (smoking, upper lobes), Panacinar (α_1 -AT deficiency, lower lobes).

Clinical Features

- **"Pink Puffers" (Emphysema-predominant):** Dyspnea, barrel chest, pursed-lip breathing, thin, hyperventilating, cachectic. Minimal cough.
- **"Blue Bloaters" (Chronic Bronchitis-predominant):** Productive cough, cyanosis, fluid retention (cor pulmonale), obese.
- *Signs:* Barrel chest (increased AP diameter), hyperresonant to percussion, decreased breath sounds, prolonged expiration, wheezes, Hoover's sign (paradoxical indrawing of lower ribs on inspiration).

Diagnosis & Investigations

- **Spirometry:** Required for diagnosis. Post-bronchodilator $FEV_1/FVC < 0.70$ confirms persistent airflow limitation. Severity staging (GOLD 1-4) based on FEV_1 % predicted.
- **Arterial Blood Gas (ABG):** Hypoxemia, hypercapnia (respiratory acidosis) in advanced disease.
- **CXR:** Hyperinflation, flattened diaphragm, increased retrosternal airspace, bullae, narrowed cardiac silhouette (emphysema).
- **ECG/Echo:** P-pulmonale (tall P waves in II, III, aVF), RV hypertrophy, Cor Pulmonale.

Treatment (GOLD Guidelines)

- **Smoking cessation:** The most effective intervention to slow the decline of lung function.
- **Vaccinations:** Influenza, Pneumococcal, COVID-19.
- **Pharmacotherapy (grouped by symptoms/exacerbation risk - ABCD assessment):**
 - *Group A:* SABA or SAMA (Ipratropium) as needed.
 - *Group B:* LABA or LAMA (Tiotropium) daily.
 - *Group E (High exacerbation risk):* LAMA + LABA, or LAMA + LABA + ICS. (Note: ICS is NEVER used as monotherapy in COPD).
 - *Phosphodiesterase-4 inhibitors (Roflumilast):* For severe chronic bronchitis with recurrent exacerbations.
- **Long-Term Oxygen Therapy (LTOT):** Prolongs survival in hypoxemic patients.

- OR PaO_2 between 56-59 mmHg with evidence of cor pulmonale, right heart failure, or polycythemia (Hematocrit $> 55\%$). Must be worn > 15 hours/day.

Complications

- Cor pulmonale (Right heart failure secondary to pulmonary hypertension from chronic hypoxic vasoconstriction).
- Secondary polycythemia.
- Pneumothorax (rupture of emphysematous bullae).
- Lung cancer (independent risk factor).

Important Exam Points (NEET-PG)

- **Reid Index:** Pathognomonic for Chronic Bronchitis. Normal < 0.4 .
- **Survival Benefit:** Only three things have been proven to decrease mortality in COPD: Smoking cessation, LTOT, and Lung Volume Reduction Surgery. (Drugs do NOT improve survival, they only improve symptoms/exacerbations).
- **Oxygen Therapy in Acute Exacerbation:** Give controlled O_2 (Venturi mask targeting 88 — 92%). Giving high O_2 can abolish the hypoxic drive, worsen V/Q mismatch (Haldane effect), and cause fatal hypercapnia.

5. HEART FAILURE (HF)

Definition

A complex clinical syndrome that results from any structural or functional impairment of ventricular filling or ejection of blood. It is defined by cardinal symptoms (dyspnea, fatigue) and signs (edema, rales) due to elevated intracardiac pressures and/or inadequate cardiac output at rest or during exercise.

Etiology

- **Ischemic Heart Disease/CAD (Most common):** Myocardial infarction causing scar/loss of functioning myocardium.
- **Hypertension:** Causes LV hypertrophy and eventual dysfunction.
- **Valvular Heart Disease:** Aortic stenosis/regurgitation, Mitral regurgitation.
- **Cardiomyopathies:** Dilated (alcohol, viral, doxorubicin, peripartum), Hypertrophic, Restrictive (Amyloidosis, Hemochromatosis).
- **High-Output HF:** Thyrotoxicosis, severe anemia, AV fistulas, Beriberi, Paget's disease.

Pathophysiology

- Index event decreases Cardiac Output (CO).
- **Neurohormonal Activation (Compensatory mechanisms that become maladaptive):**

→ increases afterload and myocardial oxygen demand → toxic to myocytes.

- **RAAS Activation:** Decreased renal perfusion → Renin release → Angiotensin II (potent vasoconstrictor, causes remodeling/fibrosis) → Aldosterone (sodium and water retention, myocardial fibrosis).
- **Natriuretic Peptides (ANP/BNP):** Released in response to atrial/ventricular stretch. Promote vasodilation and natriuresis (counter-regulatory).

Classification

- **HFrEF (Heart Failure with reduced Ejection Fraction):** $LVEF \leq 40\%$. Systolic dysfunction (inability to pump).
- **HFmrEF (Mildly reduced EF):** $LVEF 41 - 49\%$.
- **HFpEF (Heart Failure with preserved Ejection Fraction):** $LVEF \geq 50\%$. Diastolic dysfunction (inability to relax/fill). Common in elderly females with HTN and LVH.

Clinical Features

- **Left-Sided HF:** Dyspnea on exertion, Orthopnea (dyspnea when supine), Paroxysmal Nocturnal Dyspnea (PND), fatigue. Signs: Basal pulmonary crackles/rales, $S3$ gallop (volume overload), laterally displaced apex beat.
- **Right-Sided HF (Most commonly caused by Left-sided HF):** Peripheral pitting edema, ascites, right upper quadrant pain (hepatic congestion). Signs: Elevated Jugular Venous Pressure (JVP), hepatojugular reflux, tender hepatomegaly.

Diagnosis & Investigations

- **Echocardiography (Gold Standard):** Assesses LVEF, chamber sizes, wall thickness, valvular function, and diastolic function.
- **BNP or NT-proBNP:** Highly sensitive. Excellent for ruling out HF in acute dyspnea (normal levels make HF very unlikely).
- **CXR:** Cardiomegaly (Cardiothoracic ratio > 0.5), Alveolar edema (Bat's wing appearance), Kerley B lines (interstitial edema), Pleural effusion (usually right-sided or bilateral), prominent upper lobe vessels (cephalization).
- **ECG:** Evidence of LVH, prior MI (Q waves), arrhythmias (Atrial Fibrillation).

Treatment (HFrEF)

- **The "Fantastic Four" (Provide mortality benefit - Foundational therapy):**
 1. **ARNI (Sacubitril/Valsartan)** or ACEi/ARB: ARNI is superior.
 2. **Beta-Blockers:** Bisoprolol, Carvedilol, Metoprolol succinate. (Start low, go slow. Do NOT start in acute decompensated state).
 3. **Mineralocorticoid Receptor Antagonists (MRA):** Spironolactone, Eplerenone (monitor for hyperkalemia).

- **Symptom Relief:** Loop diuretics (Furosemide, Torsemide). Reduce preload/congestion. (No mortality benefit).
- **Devices:** ICD (Implantable Cardioverter-Defibrillator) for $EF \leq 35\%$ to prevent sudden cardiac death. CRT (Cardiac Resynchronization Therapy / Biventricular pacing) for $EF \leq 35\%$ with LBBB and $QRS > 150$ ms.

Complications

- Arrhythmias (Atrial fibrillation, Ventricular tachycardia/fibrillation leading to Sudden Cardiac Death).
- Cardiorenal syndrome.
- Hepatic congestion (Cardiac cirrhosis).

Important Exam Points (NEET-PG)

- **Drugs that Decrease Mortality in HFrEF:** ACEi/ARBs, ARNI, Beta-blockers, Spironolactone, SGLT2i, Hydralazine+Nitrates (in African descent).
- **Drugs that DO NOT decrease mortality:** Loop diuretics, Digoxin.
- **Digoxin:** Positive inotrope, negative chronotrope. Used for symptom control and in HF with Atrial Fibrillation. Toxicity worsened by hypokalemia. Signs of toxicity: Nausea, vomiting, blurry yellow vision (xanthopsia), arrhythmias (PVCs most common, Bidirectional VT is characteristic).
- **Heart Failure Cells:** Hemosiderin-laden macrophages in the lungs.

6. MYOCARDIAL INFARCTION (MI / ACS)

Definition

Acute Coronary Syndrome (ACS) encompasses a spectrum of conditions compatible with acute myocardial ischemia and/or infarction, due to an abrupt reduction in coronary blood flow. It includes Unstable Angina (UA), Non-ST-Segment Elevation MI (NSTEMI), and ST-Segment Elevation MI (STEMI).
MI Definition: Detection of a rise and/or fall of cardiac biomarkers (Troponin) with at least one value above the 99th percentile URL, along with evidence of myocardial ischemia (symptoms, ECG changes, or imaging).

Etiology & Pathophysiology

- **Atherosclerosis:** Rupture, fissuring, or erosion of an unstable atherosclerotic plaque → exposure of subendothelial collagen and tissue factor → platelet adhesion and aggregation → activation of coagulation cascade → Thrombus formation.
 - *Subtotal occlusion:* UA or NSTEMI.
 - *Total occlusion:* STEMI.
- **Type 2 MI:** Ischemia due to increased demand or decreased supply in the absence of acute plaque rupture (e.g., severe anemia, sepsis, coronary spasm / Prinzmetal angina).

- **Chest Pain:** Retrosternal, crushing/squeezing heavy pain. Radiates to the left arm, neck, jaw, or back. Usually lasts > 20 minutes. Not relieved by rest or nitroglycerin.
- **Autonomic symptoms:** Profuse diaphoresis (sweating), nausea, vomiting, apprehension (feeling of impending doom).
- *Atypical Presentation:* Elderly, women, and diabetics may present without chest pain (silent MI), primarily with dyspnea, fatigue, or syncope.

Diagnosis & Investigations

- **12-Lead ECG (Perform within 10 minutes):**
 - **STEMI:** ST-segment elevation ≥ 1 mm in ≥ 2 contiguous leads (except V2-V3, which have age/sex specific cutoffs: ≥ 2 mm in men ≥ 40 , ≥ 2.5 mm in men < 40 , ≥ 1.5 mm in women). Followed by T wave inversion and pathological Q waves. New LBBB is considered equivalent to STEMI in the right clinical context.
 - *Localizing the infarct:*
 - Anterior: V1-V4 (LAD territory).
 - Lateral: I, aVL, V5-V6 (LCx territory).
 - Inferior: II, III, aVF (RCA territory, sometimes LCx).
 - **NSTEMI/UA:** ST depression or T wave inversion.
- **Cardiac Biomarkers:**
 - *High-Sensitivity Troponin (I or T):* Most sensitive and specific. Rises in 3-4 hours, peaks at 24 hours, stays elevated for 7-14 days.
 - *CK-MB:* Rises in 4-6 hours, returns to normal in 48-72 hours. Useful for detecting **reinfarction**.
- **Echocardiography:** Regional wall motion abnormalities (RWMA).

Treatment

- **Immediate (MONA-B is historical; currently focus on anti-ischemic and antithrombotic):**
 - **Oxygen** (only if $SpO_2 < 90\%$).
 - **Nitrates** (Sublingual GTN - avoids in Inferior MI/RV infarct as they are preload dependent).
 - **Aspirin** (300 mg chewed).
 - **P2Y12 Inhibitor:** Ticagrelor or Prasugrel (preferred over Clopidogrel).
 - **Beta-blocker** (Oral, to reduce myocardial work; contraindicated in acute HF, heart block, or cocaine use).
 - High-intensity **Statin** (Atorvastatin 80 mg).
 - Anticoagulation: Heparin (UFH or LMWH).
- **Reperfusion Therapy (Crucial for STEMI):**

time < 90 minutes (or < 120 minutes if transferred from a non-PCI center).

- **Thrombolysis/Fibrinolysis (Tenecteplase, Alteplase):** Used if PCI is unavailable within 120 minutes. Goal: Door-to-needle time < 30 minutes. (Contraindicated in active bleeding, recent surgery, prior hemorrhagic stroke).
- **NSTEMI/UA Management:** Antithrombotics + early invasive strategy (angiography within 24-72 hours based on risk scoring e.g., GRACE/TIMI score). No thrombolysis in NSTEMI!

Complications

- **Arrhythmias:** Ventricular Fibrillation (most common cause of death in the first 48 hours).
- **Mechanical (3-14 days post-MI):** Free wall rupture (cardiac tamponade), Ventricular septal rupture (harsh holosystolic murmur), Papillary muscle rupture (acute severe mitral regurgitation - common in inferior MI).
- **Inflammatory:**
 - *Fibrinous Pericarditis:* 1-3 days post-MI.
 - *Dressler Syndrome:* 2-8 weeks post-MI (autoimmune fibrinous pericarditis).

Important Exam Points (NEET-PG)

- **Inferior MI (Right Ventricular Infarction):** Suspect if elevated JVP, clear lungs, hypotension. **Nitrates and Diuretics are strictly contraindicated.** Treatment is IV fluids to maintain preload.
- **Marker for Reinfarction:** CK-MB.
- **ECG leads:** II, III, aVF (Inferior - RCA); V1-V4 (Anterior - LAD); I, aVL, V5, V6 (Lateral - LCx).

7. PNEUMONIA

Definition

An acute infection of the pulmonary parenchyma. Classified as:

- **Community-Acquired Pneumonia (CAP):** Acquired outside of a hospital setting or diagnosed within 48 hours of admission.
- **Hospital-Acquired Pneumonia (HAP):** Occurs $\geq$ 48 hours after admission.
- **Ventilator-Associated Pneumonia (VAP):** Occurs $\geq$ 48 hours after endotracheal intubation.

Etiology

- **Typical CAP:** *Streptococcus pneumoniae* (Most common overall), *Haemophilus influenzae*, *Moraxella catarrhalis*, *Staphylococcus aureus* (post-viral/influenza).
- **Atypical CAP:** *Mycoplasma pneumoniae* (young adults, dorms), *Chlamydophila pneumoniae*, *Legionella pneumophila* (air conditioning, water tanks, associated with hyponatremia and diarrhea).
- **HAP/VAP:** Enteric Gram-negative bacilli (*Klebsiella*, *E. coli*), *Pseudomonas aeruginosa*, MRSA.

epileptics, stroke patients (typically affects dependent lobes: superior segment of lower lobe or posterior segment of upper lobe).

- **Viral:** Influenza, SARS-CoV-2, RSV.

Pathophysiology

Microorganisms reach the lower respiratory tract via microaspiration, inhalation, or hematogenous spread. Overwhelms alveolar macrophages → acute inflammatory response (cytokines, neutrophils) → exudate fills alveoli (consolidation) → V/Q mismatch and hypoxemia. *Stages of Lobar Pneumonia:* Congestion (day 1-2) → Red Hepatization (day 3-4, exudate with RBCs/fibrin) → Gray Hepatization (day 5-7, RBCs disintegrate, neutrophils/fibrin remain) → Resolution (day 8+).

Clinical Features

- **Typical:** Abrupt onset of high fever with chills/rigors, productive cough (purulent or rust-colored sputum), pleuritic chest pain, dyspnea.
- **Atypical:** Insidious onset, low-grade fever, dry cough, prominent extrapulmonary symptoms (headache, myalgia, fatigue).
- **Physical Exam:** Signs of consolidation: Tachypnea, dullness to percussion, increased tactile vocal fremitus, bronchial breath sounds, late inspiratory crackles (crepitations), egophony (E to A change), whispering pectoriloquy.

Diagnosis & Investigations

- **Chest X-Ray (CXR):** Confirms the diagnosis. Shows lobar consolidation, patchy infiltrates (bronchopneumonia), or interstitial infiltrates (atypical). Air bronchograms are a classic sign of alveolar space filling.
- **Sputum Gram Stain & Culture:** Useful for targeting therapy (must obtain before antibiotics if possible, but do not delay treatment).
- **Blood Cultures:** Positive in ~10% of cases.
- **Urinary Antigens:** For *Legionella* and *S. pneumoniae*.
- **Biomarkers:** Procalcitonin (elevated in bacterial, low in viral).

Severity Scoring (CURB-65)

Helps determine the site of care (Outpatient vs. Inpatient vs. ICU):

- **Confusion**
- **Urea** > 7 mmol/L (BUN > 19 mg/dL)
- **Respiratory rate** ≥ 30 /min
- **Blood pressure** (Systolic < 90 or Diastolic ≤ 60)
- **Age** ≥ 65 years (Score 0-1: Outpatient; Score 2: Ward admission; Score 3+: Severe, consider ICU).

Treatment (CAP)

- **Outpatient (Comorbidities):** Oral Amoxicillin/Clavulanate + Macrolide, OR Respiratory Fluoroquinolone (Levofloxacin/Moxifloxacin).
- **Inpatient (Ward):** IV Beta-lactam (Ceftriaxone) + Macrolide (Azithromycin), OR IV Respiratory Fluoroquinolone.
- **Inpatient (ICU):** IV Beta-lactam (Ceftriaxone, Cefotaxime) + IV Azithromycin or Respiratory Fluoroquinolone. (Add Vancomycin/Linezolid if MRSA suspected; use Piperacillin-Tazobactam if *Pseudomonas* suspected).

Complications

- Parapneumonic effusion, Empyema (pus in pleural space), Lung abscess (tissue necrosis with air-fluid level, common with *S. aureus*, *Klebsiella*, Anaerobes), Sepsis, ARDS.

Important Exam Points (NEET-PG)

- ***Klebsiella pneumoniae*:** "Currant jelly" sputum, bulging fissure sign on CXR, common in alcoholics.
- ***Legionella*:** Hyponatremia, diarrhea, relative bradycardia, neurological symptoms. Diagnosis by urine antigen. Treatment: Macrolides or Fluoroquinolones.
- ***Mycoplasma pneumoniae*:** Cold agglutinins (IgM leading to autoimmune hemolytic anemia), erythema multiforme.
- ***Pneumocystis jirovecii* (PCP):** Fungal pneumonia in HIV/AIDS ($CD4 < 200$). Bilateral ground-glass opacities. Treat/Prophylax with Cotrimoxazole (TMP-SMX).

8. TUBERCULOSIS (TB)

Definition

A chronic, necrotizing granulomatous infectious disease caused by *Mycobacterium tuberculosis* complex, primarily affecting the lungs (Pulmonary TB) but can affect any organ (Extrapulmonary TB).

Etiology

- *Mycobacterium tuberculosis* is an obligate aerobe, non-motile, non-spore-forming bacillus.
- Has a lipid-rich cell wall containing mycolic acid, making it acid-fast (retains carbol-fuchsin stain despite acid-alcohol wash).

Pathophysiology

- **Primary TB:** Inhalation of droplet nuclei → Bacilli reach alveoli → Phagocytosed by alveolar macrophages but survive intracellularly by inhibiting phagolysosome fusion → Multiplication → Cell-mediated immunity (Type IV hypersensitivity) develops after 2-4 weeks → $CD4^+$ T cells secrete $IFN-\gamma$ → Macrophage activation and differentiation into epithelioid cells and Langhans giant cells → Formation of **caseating granulomas**.
 - *Ghon focus*: Subpleural lesion in mid/lower zones.

- *Ranke complex*: Calcified Ghon complex.
- **Secondary (Reactivation) TB**: Reactivation of latent infection, usually when host immunity drops (HIV, diabetes, steroids, aging). Typically occurs in the apices of the lungs (high oxygen tension). Tends to cavitate.

Clinical Features

- **Pulmonary TB**: Chronic cough (dry initially, then purulent), Hemoptysis, Low-grade evening-rise fever, Night sweats, Anorexia, Weight loss.
- **Extrapulmonary TB (EPTB)**:
 - *Lymphadenitis*: Most common EPTB. Painless, matted cervical nodes (Scrofula).
 - *Pleural TB*: Exudative effusion with high lymphocytes and high ADA (Adenosine Deaminase).
 - *Skeletal TB*: Pott's disease (TB of the spine), cold abscesses.
 - *CNS TB*: TB meningitis (basal exudates, cranial nerve palsies, CSF shows high protein, low glucose, lymphocytosis).
 - *Genitourinary*: "Sterile" pyuria, epididymitis, female infertility (endometrial TB).

Diagnosis & Investigations

- **Sputum Smear Microscopy**: Ziehl-Neelsen (ZN) stain or Fluorescent staining (Auramine-Rhodamine). Requires at least 2 samples (one early morning).
- **CBNAAT (GeneXpert MTB/RIF) / Truenat**: Nucleic acid amplification test. **Gold standard for rapid diagnosis**. Detects TB DNA and Rifampicin resistance within 2 hours. Endorsed by NTEP as the initial diagnostic test.
- **Culture**: MGIT (Mycobacteria Growth Indicator Tube) - liquid culture is faster (1-3 weeks) than solid Lowenstein-Jensen (LJ) medium (4-8 weeks). Gold standard for drug susceptibility testing.
- **CXR**: Apical cavitations, fibronodular infiltrates, miliary mottling (millet seed appearance).
- **Latent TB Tests**: Mantoux test (Tuberculin Skin Test - measures delayed hypersensitivity) and IGRA (Interferon-Gamma Release Assay - e.g., QuantiFERON-TB Gold). These do not distinguish latent from active disease.

Treatment (NTEP Guidelines)

- **First-Line Drugs (HRZE)**:
 - **Isoniazid (H)**: Bactericidal. SE: Peripheral neuropathy (prevent with Pyridoxine/B6), Hepatitis, Drug-induced lupus.
 - **Rifampicin (R)**: Bactericidal. SE: Orange-red urine/secretions, Hepatitis, potent CYP450 inducer (decreases efficacy of OCPs, ART).
 - **Pyrazinamide (Z)**: Acts in acidic environment (intracellular). SE: Hyperuricemia (gout), most hepatotoxic of all.
 - **Ethambutol (E)**: Bacteriostatic. SE: Optic neuritis (decreased visual acuity, red-green color blindness). NOT hepatotoxic.

- *Intensive Phase*: 2 months of HRZE.
- *Continuation Phase*: 4 months of HRE (some guidelines use HR). Total duration: 6 months.

Complications

- Massive hemoptysis (rupture of Rasmussen's aneurysm).
- Spontaneous pneumothorax, bronchiectasis.
- Aspergilloma (fungus ball forming in a pre-existing healed TB cavity). Cor Pulmonale.

Important Exam Points (NEET-PG)

- **MDR-TB**: Resistant to at least Isoniazid and Rifampicin.
- **XDR-TB**: MDR-TB + resistant to any fluoroquinolone and at least one additional Group A drug (Bedaquiline or Linezolid).
- **Most Hepatotoxic Anti-TB Drug**: Pyrazinamide > Isoniazid > Rifampicin. Ethambutol and Streptomycin are safe for the liver.
- **Safe in Pregnancy**: H, R, Z, E. (Streptomycin is absolutely contraindicated - causes fetal ototoxicity).

9. ANEMIA

Definition

A reduction in the total circulating red cell mass, clinically indicated by a decrease in Hemoglobin (Hb) concentration or Hematocrit below the lower limit of normal for age, sex, and altitude. (WHO criteria: Hb < 13.0 g/dL in adult men, < 12.0 g/dL in non-pregnant adult women, < 11.0 g/dL in pregnancy).

Classification based on Mean Corpuscular Volume (MCV)

1. **Microcytic Hypochromic (MCV < 80 fL)**: TAILS (Thalassemia, Anemia of chronic disease, Iron deficiency, Lead poisoning, Sideroblastic anemia).
2. **Macrocytic (MCV > 100 fL)**:
 - *Megaloblastic*: Vitamin B12 deficiency, Folate deficiency. (Impaired DNA synthesis).
 - *Non-Megaloblastic*: Liver disease, Alcoholism, Hypothyroidism, Reticulocytosis.
3. **Normocytic Normochromic (MCV 80-100 fL)**: Acute blood loss, Hemolysis, Chronic kidney disease (low EPO), Aplastic anemia.

A. Iron Deficiency Anemia (IDA) - Most Common Worldwide

- **Etiology**: Chronic blood loss (GI bleed, heavy menses, hookworm in developing nations), increased demand (pregnancy, childhood), decreased absorption (Celiac disease, post-gastrectomy).
- **Pathophysiology**: Depletion of iron stores (Ferritin) → Lack of iron for heme synthesis → Microcytic hypochromic RBCs.

shaped nails), angular cheilitis, glossitis. Plummer-Vinson syndrome (Triad of IDA, esophageal web, dysphagia).

- **Diagnosis:** Hb (low), MCV (low), MCH (low). Peripheral smear: Microcytic hypochromic cells, pencil/cigar cells.
 - *Iron Profile:* Serum Ferritin (Low - most sensitive), Serum Iron (Low), TIBC (High), Transferrin saturation (Low < 15%).
- **Treatment:** Identify and treat the underlying cause. Oral elemental iron (Ferrous sulfate/fumarate/ascorbate). IV iron if oral is intolerant or malabsorption is present. *Reticulocyte count increases in 5-7 days; Hb normalizes in 2 months.*

B. Megaloblastic Anemia (B12 / Folate Deficiency)

- **Etiology:**
 - *Vit B12 (Cobalamin):* Pernicious anemia (autoantibodies against intrinsic factor or parietal cells), strict vegan diet, Crohn's disease, terminal ileum resection, *Diphyllobothrium latum* (fish tapeworm).
 - *Folate (B9):* Dietary lack (tea and toast diet), alcoholism, increased demand (pregnancy), drugs (Methotrexate, Phenytoin).
- **Pathophysiology:** Defective DNA synthesis causing nuclear-cytoplasmic asynchrony.
- **Clinical Features:** Pallor, mild jaundice (ineffective erythropoiesis), beefy red tongue.
 - *Specific to B12: Subacute Combined Degeneration of Spinal Cord* (demyelination of dorsal columns and lateral corticospinal tracts) causing loss of position/vibration sense, ataxia, upper motor neuron signs. Folate deficiency has NO neurological symptoms.
- **Diagnosis:** High MCV. Peripheral smear: Macro-ovalocytes, **hypersegmented neutrophils** (≥ 5 lobes).
 - Homocysteine is elevated in both.
 - Methylmalonic acid (MMA) is elevated ONLY in B12 deficiency.
- **Treatment:** Parenteral/oral B12, Oral Folic acid. *Caution: Treating B12 deficiency with folate alone corrects the anemia but allows neurological damage to progress irreversibly.*

C. Hemolytic Anemias

- **Features:** Jaundice (unconjugated bilirubinemia), gallstones (pigment stones), splenomegaly, dark urine. Lab: High reticulocyte count, high LDH, low Haptoglobin.
- **Sickle Cell Anemia:** Point mutation (Glutamic acid to Valine at 6th position of β -globin chain). Causes HbS polymerization under hypoxia. Presents with vaso-occlusive crises (pain, dactylitis), autosplenectomy (risk of encapsulated organism infections). Diagnosis: Hb electrophoresis. Treatment: Hydroxyurea (increases HbF).
- **Thalassemia:** Quantitative defect in globin chain synthesis. β -Thalassemia major presents with severe anemia at 6 months of age (when HbF transitions to HbA), hepatosplenomegaly, "crew-cut" appearance on skull X-ray, chipmunk facies. Diagnosis: Hb electrophoresis (elevated HbA2 and HbF).

- *Warm (IgG)*: Extravascular hemolysis in spleen. SLE, CLL, drugs. Direct Coombs test positive.
- *Cold (IgM)*: Intravascular hemolysis. Mycoplasma, EBV.
- **G6PD Deficiency**: X-linked recessive. Oxidative stress (Sulfa drugs, Antimalarials, Fava beans, infections) causes hemolysis. Smear: Bite cells, Heinz bodies.

Important Exam Points (NEET-PG)

- **Mentzer Index**: MCV / RBC count. Mentzer Index > 13 suggests IDA; < 13 suggests Thalassemia trait.
- **Pancytopenia with low reticulocyte count**: Think Aplastic Anemia (bone marrow hypocellularity, fat replacement).
- **Schistocytes (Helmet cells)**: Microangiopathic hemolytic anemia (TTP, HUS, DIC).

10. THYROID DISORDERS

Definition

Disorders involving abnormal production of thyroid hormones (T3 and T4), leading to a systemic state of decreased metabolic rate (Hypothyroidism) or increased metabolic rate (Hyperthyroidism/Thyrotoxicosis).

A. HYPOTHYROIDISM

- **Etiology**:
 - *Primary (Most common)*: Hashimoto's thyroiditis (autoimmune), Iatrogenic (post-radioactive iodine or thyroidectomy), Iodine deficiency (most common cause worldwide).
 - *Secondary/Tertiary*: Pituitary adenoma, Hypothalamic disease (rare).
- **Pathophysiology**: Hashimoto's involves $CD8^+$ T-cell mediated destruction of thyroid follicles and autoantibodies. Lack of T3/T4 decreases basal metabolic rate and increases deposition of glycosaminoglycans (myxedema).
- **Clinical Features**: Fatigue, cold intolerance, weight gain despite poor appetite, constipation, dry thick skin, hair loss (outer third of eyebrows), bradycardia, delayed relaxation phase of deep tendon reflexes, hoarseness, menstrual irregularities (menorrhagia), depressed mood.
- **Diagnosis**:
 - *Primary*: High TSH, Low Free T4. (Subclinical: High TSH, Normal Free T4).
 - *Secondary*: Low/Normal TSH, Low Free T4.
 - *Antibodies*: Anti-TPO (Thyroid Peroxidase) and Anti-Thyroglobulin antibodies are massively elevated in Hashimoto's.
- **Treatment**: Levothyroxine (synthetic T4) once daily on an empty stomach. Monitor TSH levels every 6-8 weeks to adjust the dose. Target TSH is typically 0.5-4.0 mIU/L (tighter control in pregnancy).

respiratory failure, altered sensorium). Treat with IV Levothyroxine, IV Hydrocortisone, warming.

B. HYPERTHYROIDISM (THYROTOXICOSIS)

- **Etiology:**
 - **Graves' Disease (Most common, 60-80%):** Autoimmune disorder.
 - **Toxic Multinodular Goiter (TMNG):** Autonomous functioning thyroid nodules (usually older adults).
 - **Toxic Adenoma:** Single hyperfunctioning nodule.
 - **Thyroiditis:** Subacute (De Quervain's - painful, post-viral), Silent/Postpartum. Initial destruction releases pre-formed hormone (transient hyperthyroidism).
- **Pathophysiology of Graves':** IgG autoantibodies bind to and stimulate the TSH receptor (Thyroid Stimulating Immunoglobulins - TSI). Drives excess T3/T4 production and diffuse glandular growth.
- **Clinical Features:** Heat intolerance, weight loss despite increased appetite, palpitations, tremors, sweating, frequent bowel movements, anxiety, tachycardia, atrial fibrillation.
 - *Specific to Graves' (Extrathyroidal manifestations due to TSI action on fibroblasts):* Exophthalmos/Proptosis (bulging eyes), Pretibial myxedema (non-pitting edema on shins), Thyroid acropachy (clubbing).
- **Diagnosis:**
 - Low/suppressed TSH (< 0.1), High Free T4 and/or T3.
 - *Antibodies:* TSI / TRAB (TSH Receptor Antibodies) positive in Graves'.
 - *Radioactive Iodine Uptake (RAIU) Scan:*
 - Diffuse high uptake: Graves'.
 - Patchy multiple areas of high uptake: TMNG.
 - Single hot nodule: Toxic adenoma.
 - Low/absent uptake: Thyroiditis or exogenous hormone intake (factitious).
- **Treatment:**
 - *Symptom control:* Beta-blockers (Propranolol - also blocks peripheral conversion of T4 to T3).
 - *Antithyroid Drugs (Thionamides):*
 - **Methimazole / Carbimazole:** First-line for most.
 - **Propylthiouracil (PTU):** Preferred in the *first trimester of pregnancy* (Methimazole causes aplasia cutis) and in Thyroid Storm. Inhibits peripheral conversion of T4 to T3. Major side effect for both: **Agranulocytosis** (presents as fever and sore throat), Hepatotoxicity.
 - *Radioactive Iodine Ablation (^{131}I):* Definitive therapy. Can worsen Graves' ophthalmopathy.
 - *Surgery:* Subtotal/total thyroidectomy (risk of recurrent laryngeal nerve injury, hypoparathyroidism).
- **Complication:** Thyroid Storm. Acute, life-threatening hypermetabolic state (fever $> 104^{\circ}F$, delirium, tachycardia, shock). Treat with Propranolol, PTU, SSKI (Lugol's iodine - given 1 hour *after*

Important Exam Points (NEET PG)